

IdxAi.

AI-Powered Thematic Investment Intelligence

Amazon Bedrock · Claude 3 · Generative AI · Thematic Indexing · Agentic Finance

6–9 Months

→ Few Hours

\$M Cost

→ Near Zero

300+

Thematic Baskets

Bedrock

Core Engine

EXECUTIVE SUMMARY

What is IndexGPT?

IndexGPT is JP Morgan Chase's generative AI-powered thematic investment platform, unveiled in 2023–24 to automate the creation of investment baskets based on emerging financial trends. Re-architected to leverage Amazon Bedrock as the managed AI layer, replacing direct OpenAI API calls with enterprise-grade, secure, scalable foundation model access.

Key Highlights:

- Core Engine: Amazon Bedrock (Claude 3 Sonnet / Titan)
- Trademark filed: May 2023 (USPTO)
- Product type: Agentic, multi-stage AI pipeline
- Target: Institutional → Retail investors
- First major bank to release GPT-class product to customers
- Bedrock provides serverless, IAM-secured, multi-model access

Source: CNBC (May 2023) · USPTO Filing · LinkedIn Case Study (2024)

Speed

6–9 months → hours

Cost

Millions → Near Zero

Accuracy

2× more keywords vs legacy tools

Real-Time

Always-on, autonomous rebalancing

Coverage

Equities, Bonds, ESG, Crypto

PROBLEM STATEMENT

Why JPMorgan needed to reinvent thematic investing

01

Months-Long Index Creation

Building a single thematic investment basket required 6–9 months of manual research by teams of analysts — making niche, fast-moving themes commercially unviable.

02

Million-Dollar Cost Per Index

Legacy processes cost millions of dollars per index, limiting portfolio customization only to large institutional mandates and blocking retail democratization.

03

Static, Backwards-Looking Portfolios

Traditional sector indices lagged real-world trends. By the time an 'e-sports' or 'cloud computing' index was ready, the market opportunity had already partially played out.

04

No Autonomous Analyst Coverage

Human analysts couldn't continuously screen thousands of companies across news, SEC filings, and earnings — creating blind spots in theme tracking and security selection.

SOLUTION OVERVIEW

IndexGPT: A Multi-Stage Agentic AI Pipeline powered by Amazon Bedrock

Robust & Scalable Architecture

- Primary AI Core: Amazon Bedrock — Claude 3 Sonnet for theme generation & keyword extraction
- Secondary Engine: Amazon Titan Embeddings for high-volume news & semantic scanning
- Architecture Style: Agentic AI — autonomous task execution without human-in-the-loop
- Infrastructure: AWS-native, elastic compute via SageMaker & Lambda for real-time processing
- Data Sources: Market data, news articles, SEC filings, economic indicators, corporate reports
- Asset Coverage: Equities, Fixed Income, ESG, Cryptocurrencies
- Security: IAM roles, AWS PrivateLink — no data leaves JPMorgan's VPC

Thematic Indexing

Niche trend-based baskets (e.g. 'Remote Work' not just 'Tech') via Bedrock Claude 3

Personalization

Tailored portfolios by risk tolerance, goals & preferred themes

Real-Time Adaptation

Continuous SEC filing & news screening — always-on analyst via Bedrock streaming

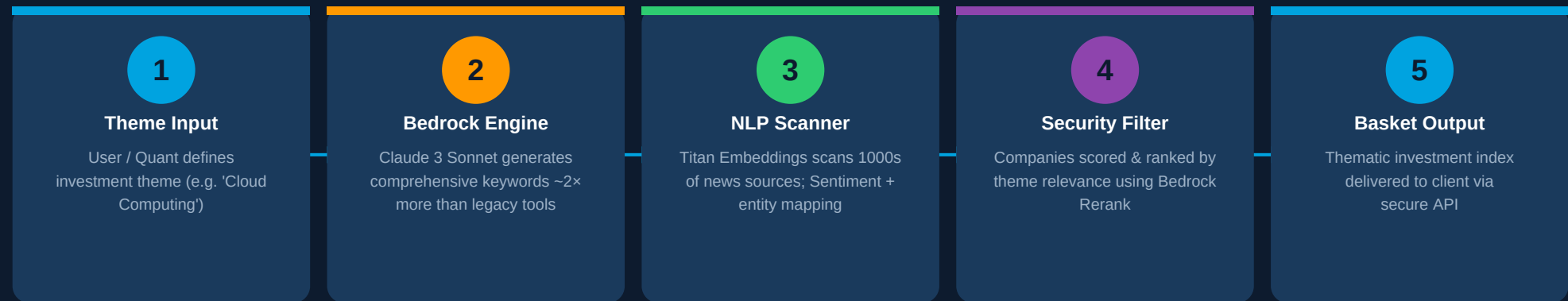
Democratization

Reduces cost to near zero, enabling retail investor access

Powered by Amazon Bedrock | Claude 3 Sonnet + Titan Embeddings

TECHNICAL ARCHITECTURE

Multi-Stage Agentic Pipeline — From Theme to Investment Basket



Amazon Bedrock (Claude 3)

Managed LLM access — theme gen & keyword extraction. Produces 2× keyword density vs prior software. No infra to manage.

Amazon Titan Embeddings

Secondary model for high-volume article scanning & semantic entity recognition at scale. Integrated via Bedrock API.

AWS Cloud Infrastructure

Lambda + SageMaker + EventBridge handle real-time market data; deployed as cloud-native serverless SaaS within JPM VPC.

Agentic Autonomy

Step Functions orchestrate the pipeline; Bedrock Agents autonomously read filings, screen companies — a virtual analyst team.

TECHNICAL WORKFLOW — INTERNAL DETAILS

Step 1: Theme Generation & Keyword Extraction

- Claude 3 Sonnet (via Bedrock) receives a high-level theme (e.g. 'E-Sports', 'Genomics')
- Generates a comprehensive keyword taxonomy covering all sub-sectors
- Produces ~2× more relevant keywords than prior software solutions
- Keywords structured as a semantic graph using Bedrock Knowledge Base

Step 2: News & Sentiment Scanning

- Keywords fed into Amazon Titan Embeddings via Bedrock API
- Model scans thousands of news articles, press releases & media sources
- Identifies companies actively mentioned in context of the theme
- Sentiment scoring applied: positive/negative association weighted via Comprehend

Step 3: Security Selection & Basket Construction

- Companies filtered by relevance score, market cap & liquidity criteria
- Final basket of securities constructed — the 'thematic investment index'
- Supports equities, fixed income, ESG-focused assets & cryptocurrencies
- Output delivered as a structured index to portfolio management systems

Step 4: Real-Time Monitoring & Rebalancing

- Bedrock Agents act as 'always-on' autonomous analyst post-basket creation
- Continuously re-screens financial reports, news & earnings data via EventBridge
- Triggers rebalancing signals when company relevance scores shift
- No human-in-the-loop required for ongoing monitoring

AI MODELS & HIDDEN TECHNICAL DETAILS

Under the hood: What powers IndexGPT on Amazon Bedrock

Primary Model: Amazon Bedrock — Claude 3 Sonnet

- Model Type: Large Language Model (Anthropic Claude 3 Sonnet)
- Role: Theme understanding, keyword taxonomy generation, semantic reasoning
- Key Capability: Contextual understanding of financial themes across macro & micro domains
- Output: Structured keyword lists + semantic relationships between entities
- Advantage: Produces 2× keyword depth compared to traditional keyword tools
- Access: Managed via Bedrock API — no model weights to maintain or scale
- Security: Requests stay within AWS VPC; no training on JPM data
- Compliance: SOC 2, HIPAA-eligible, FedRAMP via AWS infrastructure

Secondary Model: Amazon Titan Embeddings

- Model Type: Purpose-built embeddings model via Amazon Bedrock
- Role: High-volume text processing AFTER Claude 3 generates keywords
- Input: Keyword taxonomy from Claude 3 + corpus of news/financial documents
- Processing: Entity recognition, semantic similarity, company mention scoring
- Scale: Scans thousands of articles & filings per theme in minutes
- Why Bedrock Titan? Optimized for bulk scan; cost-effective at high volume
- Integration: Bedrock Rerank API for relevance scoring in basket construction

DATA SOURCES, COVERAGE & SPECIFICATIONS

Data Inputs

Market Performance Data

Live price feeds, volume, market cap, momentum signals across global exchanges

News Articles & Media

Real-time scan of financial news, press releases, earnings transcripts & analyst reports

Economic Indicators

Macro signals: GDP, inflation, sector rotation triggers, central bank data

Corporate Financial Reports

SEC filings (10-K, 10-Q, 8-K), EDGAR data, international equivalents

Asset Classes & Specs

Equities

Global stocks, growth & value, sector & sub-sector

Fixed Income

Government & corporate bonds, duration strategies

ESG Assets

Environment, Social & Governance screened baskets

Cryptocurrencies

Digital assets integrated as thematic exposure

Technical Specifications

Trademark: USPTO — 'cloud computing software using AI for analyzing and selecting securities tailored to customer needs' | Deployment: AWS Cloud SaaS | Models: Bedrock Claude 3 + Titan

BUSINESS VALUE ACHIEVED

6–9 Mo
→ **Few Hours**

Index Creation Time

~\$0
vs. Millions

Cost Per Index

2×
Keyword Depth

vs Legacy Tools

24/7
Autonomous

Market Monitoring

Operational Efficiency

- Index creation compressed from months to hours — a fundamental step-change in speed-to-market
- Analysts freed from manual keyword research to focus on strategy and client advisory
- Bedrock's managed API eliminates model hosting overhead

Cost Optimization

- Index creation cost reduced from millions to near zero per thematic basket
- Enables profitability on hyper-niche indices that were previously commercially unviable
- Bedrock pay-per-token pricing aligns cost directly with usage

Scalability & Market Position

- First major bank to release a GPT-class product directly to customers
- Positions JPMorgan as lead innovator in AI-driven financial services globally
- AWS enterprise agreements provide preferred pricing & SLA guarantees

Business Growth

- Institutional clients: superior theme representation; Retail roadmap: Latin America pilot
- Continuous real-time rebalancing replaces periodic manual reviews — living portfolios
- Bedrock Agents enable future expansion to fully autonomous portfolio management

STRATEGIC CONCLUSION

IndexGPT — Redefining What's Possible in Financial AI with Amazon Bedrock

JP Morgan's IndexGPT represents the most ambitious application of Generative AI in mainstream financial services — now re-architected on Amazon Bedrock, combining Claude 3 Sonnet's reasoning with Amazon Titan Embeddings at scale to autonomously construct, monitor, and rebalance thematic investment portfolios. It does not merely automate existing processes; it makes entirely new product categories commercially viable — all within a secure, compliant AWS environment.

First Mover Advantage

First bank to patent & launch a GPT-class product to customers — 3-year trademark window creates strong moat

Democratization of Alpha

Ultra-niche themes (e.g. 'AI in Agriculture') now viable for retail — unlocking a new revenue category via Bedrock

Autonomous Finance

IndexGPT is a blueprint for agentic AI in finance — the beginning of AI-run portfolio management on AWS

Key Facts Recap

Trademark: IndexGPT (USPTO, May 2023) · Models: Bedrock Claude 3 Sonnet + Amazon Titan · Pipeline: 4-Stage Agentic AI · Cost: Near Zero per index · Time: Hours vs Months · Audiences: Institutional + Retail · Co

Sources: CNBC (May 2023) · USPTO Filing · LinkedIn Case Study (2024) · FinTech Global · Business Times · AWS Documentation